BioNix_Wesley_9_ILP

Wednesday, 10 September 2025 14:52

Try to compress multiple tools in one pkgs

nix-shell --pure -p '(rWrapper.override {packages = with rPackages; [ggplot2];})' --command R

nix-shell --pure -p python3 python3Packages.numpy --command python

```
{ pkgs ? import <nixpkgs> {} }:

pkgs.mkShell {
  buildInputs = [
    pkgs.python3
    pkgs.python3Packages.fastapi
    pkgs.python3Packages.numpy
    pkgs.python3Packages.pip  # Optional, if you need pip
  ];

  shellHook = ''
    echo "Welcome to the FastAPI and NumPy development environment!"
    exec python
  '';
}
```

Try to do those in this week

Individual Learning Plan – Week 9

Goals for Week 9

- Consolidate understanding of reproducibility environments in Nix.
- Extend the R-focused reproducibility package built in Week 8 by testing practical data analysis workflows (e.g., QQ plots).
- Begin linking reproducibility practice with research use-cases (how such an environment can support future data analysis pipelines).

Activities Undertaken

- Initialized and committed the reproducibility package into a Git repository to enable lock-file tracking and version pinning.
- Successfully entered the pinned R environment using nix develop and ran the demo script (qq_demo.R).
- Generated multiple QQ plots (qq_base.png, qq_car.png, qq_ggplot.png) from simulated normal distribution data to test that statistical visualization works inside the reproducible environment.
- Debugged common issues, including Git tracking requirements for flakes and ensuring R

Debugged common issues, including Git tracking requirements for flakes and ensuring R packages (qqplotr, car) are only called inside the Nix-managed environment.

Key Learnings

- Reproducibility requires both the environment (Nix flake) and source control (Git) to work together.
- The data in the demo was generated with rnorm(), highlighting the difference between reproducible simulation and using external datasets.
- Using set.seed() ensures reproducibility even in random data generation.
- The reproducibility package can be adapted easily to real-world data analysis by replacing the simulated dataset with imported CSV or experimental data.